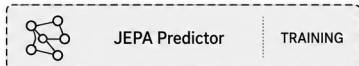
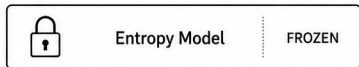
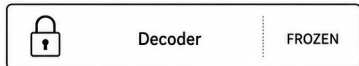
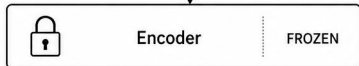


PHASE 1

Pre-training (20 epochs)

$$\text{MSE loss: } \|z_t - P(z_{<t})\|^2$$



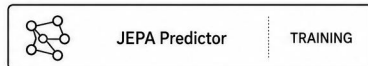
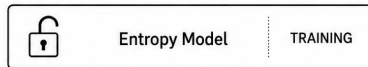
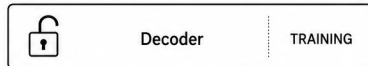
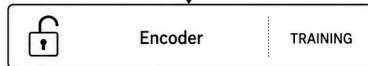
Prediction MSE: $0.226 \rightarrow 1.15 \times 10^{-4}$

θ_P initialized

PHASE 2

Joint Fine-tuning (80 epochs)

$$\text{RD loss: } \lambda R + D$$



P/I ratio: $0.93\times \rightarrow 0.37\times$